Certified Ethical Hacking & AI – Expert

120-Hours (6 months) Advanced Offensive Security, SOC, Detection Engineering & AI Security Program

Duration	Level
120 Hours (6 months)	Expert / Advanced

Course Purpose

Advanced engineering program for learners with foundational cybersecurity knowledge. The Expert track moves beyond basic ethical hacking into enterprise SOC operations, Splunk Enterprise Security, Microsoft Sentinel, detection engineering, threat hunting, SOAR, AI security agents, GenAI/LLM security and AI governance.

Module-wise Duration Summary

Module	Topic	Hours
01	Advanced Ethical Hacking Methodology & Attack Surface Strategy	5
02	Advanced Reconnaissance, OSINT & Attack Surface Management	4
03	Advanced Network Scanning & Enumeration	4
04	Advanced Vulnerability Analysis & Exploit Validation	5
05	Advanced Web Application & API Security	5
06	SQL Injection, OWASP Web Security & Secure Remediation	4
07	System & Endpoint Security Engineering	4
08	Advanced Linux Security & Hardening	3
09	Wireless & Network Security Assessment	3
10	Malware Reverse Engineering & Threat Analysis	5
11	Advanced Cryptography, PKI & Secure Communications	3
12	Network Forensics & Advanced Traffic Analysis	4
13	SOC Architecture & Security Operations	4
14	Advanced Splunk Investigation & Detection	5
15	Splunk Enterprise Security, CIM & Risk-Based Analytics	5
16	Advanced Microsoft Sentinel & KQL Investigation	5
17	SIEM Detection Engineering & Content Development	5
18	Incident Response, Threat Hunting & Investigation	5
19	Threat Intelligence & MITRE ATT&CK	4
20	Security Automation & SOAR Engineering	4
21	AI-Powered SOC Operations	4
22	AI Security Analyst & Agent Engineering	5
23	LLM / GenAI Application Security	4

24	AI-Driven Threat Detection, UEBA & Security Analytics	4
25	AI Security Architecture, Governance & Risk	3
26	Integrated Red-Team / Blue-Team Case Studies	4
27	Advanced Security Labs & Multi-Stage Attack Simulation	4
28	Expert Capstone Projects	6
	TOTAL	**120**

Module 01: Advanced Ethical Hacking Methodology & Attack Surface Strategy — 5 Hours

Theory: Advanced Ethical Hacking Methodology & Attack Surface Strategy: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Ethical Hacking Methodology & Attack Surface Strategy methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 02: Advanced Reconnaissance, OSINT & Attack Surface Management — 4 Hours

Theory: Advanced Reconnaissance, OSINT & Attack Surface Management: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Reconnaissance, OSINT & Attack Surface Management methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Amass, dig, nslookup, RDAP/WHOIS

Module 03: Advanced Network Scanning & Enumeration — 4 Hours

Theory: Advanced Network Scanning & Enumeration: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Network Scanning & Enumeration methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Nmap, Wireshark, isolated lab hosts

Module 04: Advanced Vulnerability Analysis & Exploit Validation — 5 Hours

Theory: Advanced Vulnerability Analysis & Exploit Validation: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Vulnerability Analysis & Exploit Validation methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Greenbone/OpenVAS, Nuclei, controlled lab targets

Module 05: Advanced Web Application & API Security — 5 Hours

Theory: Advanced Web Application & API Security: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Web Application & API Security methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: OWASP ZAP, Burp Community, Juice Shop/WebGoat/DVWA

Module 06: SQL Injection, OWASP Web Security & Secure Remediation — 4 Hours

Theory: SQL Injection, OWASP Web Security & Secure Remediation: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the SQL Injection, OWASP Web Security & Secure Remediation methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: OWASP ZAP, Burp Community, Juice Shop/WebGoat/DVWA

Module 07: System & Endpoint Security Engineering — 4 Hours

Theory: System & Endpoint Security Engineering: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the System & Endpoint Security Engineering methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 08: Advanced Linux Security & Hardening — 3 Hours

Theory: Advanced Linux Security & Hardening: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Linux Security & Hardening methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, Ubuntu, auditd, UFW/nftables

Module 09: Wireless & Network Security Assessment — 3 Hours

Theory: Wireless & Network Security Assessment: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Wireless & Network Security Assessment methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kismet, Aircrack-ng, Wireshark, isolated instructor-owned AP

Module 10: Malware Reverse Engineering & Threat Analysis — 5 Hours

Theory: Malware Reverse Engineering & Threat Analysis: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Malware Reverse Engineering & Threat Analysis methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Ghidra, YARA, ClamAV, safe training samples

Module 11: Advanced Cryptography, PKI & Secure Communications — 3 Hours

Theory: Advanced Cryptography, PKI & Secure Communications: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Cryptography, PKI & Secure Communications methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: OpenSSL, GnuPG, CyberChef, Python

Module 12: Network Forensics & Advanced Traffic Analysis — 4 Hours

Theory: Network Forensics & Advanced Traffic Analysis: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Network Forensics & Advanced Traffic Analysis methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Wireshark, tcpdump, Zeek, provided PCAPs

Module 13: SOC Architecture & Security Operations — 4 Hours

Theory: SOC Architecture & Security Operations: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the SOC Architecture & Security Operations methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 14: Advanced Splunk Investigation & Detection — 5 Hours

Theory: Advanced SPL, correlation, enrichment, hunting and detection development.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Splunk Investigation & Detection methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Splunk Enterprise / Enterprise Security where available, synthetic logs

Module 15: Splunk Enterprise Security, CIM & Risk-Based Analytics — 5 Hours

Theory: CIM, data models, findings, Mission Control concepts and risk-based analytics.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Splunk Enterprise Security, CIM & Risk-Based Analytics methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Splunk Enterprise / Enterprise Security where available, synthetic logs

Module 16: Advanced Microsoft Sentinel & KQL Investigation — 5 Hours

Theory: Advanced KQL, entities, analytics rules, hunting and incident investigation.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Microsoft Sentinel & KQL Investigation methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Microsoft Sentinel, Log Analytics, KQL, synthetic telemetry

Module 17: SIEM Detection Engineering & Content Development — 5 Hours

Theory: Detection lifecycle, tuning, testing, ATT&CK mapping and detection-as-code.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the SIEM Detection Engineering & Content Development methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 18: Incident Response, Threat Hunting & Investigation — 5 Hours

Theory: Incident Response, Threat Hunting & Investigation: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Incident Response, Threat Hunting & Investigation methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 19: Threat Intelligence & MITRE ATT&CK — 4 Hours

Theory: Threat Intelligence & MITRE ATT&CK: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Threat Intelligence & MITRE ATT&CK methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 20: Security Automation & SOAR Engineering — 4 Hours

Theory: Playbooks, approvals, API enrichment, case management and rollback.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Security Automation & SOAR Engineering methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Shuffle or n8n, Python, local APIs

Module 21: AI-Powered SOC Operations — 4 Hours

Theory: AI-Powered SOC Operations: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the AI-Powered SOC Operations methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Ollama, Python, local datasets

Module 22: AI Security Analyst & Agent Engineering — 5 Hours

Theory: Agent architecture, tools, state, planning, verification, guardrails and prompt injection.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the AI Security Analyst & Agent Engineering methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Ollama, Python, local datasets

Module 23: LLM / GenAI Application Security — 4 Hours

Theory: Prompt injection, insecure output, data leakage, supply chain, excessive agency and RAG risks.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the LLM / GenAI Application Security methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Ollama, Python, local datasets

Module 24: AI-Driven Threat Detection, UEBA & Security Analytics — 4 Hours

Theory: Behavioral analytics, anomaly detection, risk scoring, explainability and validation.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the AI-Driven Threat Detection, UEBA & Security Analytics methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Ollama, Python, local datasets

Module 25: AI Security Architecture, Governance & Risk — 3 Hours

Theory: AI lifecycle security, threat modeling, governance, privacy and secure deployment.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the AI Security Architecture, Governance & Risk methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Ollama, Python, local datasets

Module 26: Integrated Red-Team / Blue-Team Case Studies — 4 Hours

Theory: Integrated Red-Team / Blue-Team Case Studies: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Integrated Red-Team / Blue-Team Case Studies methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 27: Advanced Security Labs & Multi-Stage Attack Simulation — 4 Hours

Theory: Advanced Security Labs & Multi-Stage Attack Simulation: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Advanced Security Labs & Multi-Stage Attack Simulation methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Module 28: Expert Capstone Projects — 6 Hours

Theory: Expert Capstone Projects: concepts, methodology, common risks, defensive controls, evidence and professional reporting.

Practical Lab: Hands-on exercise in an isolated authorized lab: apply the Expert Capstone Projects methodology to deliberately vulnerable systems or synthetic security data, collect evidence, validate results and document remediation.

Tools: Kali Linux, VirtualBox, Python and controlled lab datasets

Free / Open-Source Lab Stack

- Kali Linux, Ubuntu and VirtualBox for isolated labs.
- Nmap, Wireshark, OWASP ZAP, Burp Suite Community, Juice Shop/WebGoat/DVWA.
- Greenbone/OpenVAS, Nuclei, John the Ripper, Hashcat, OpenSSL, GnuPG, CyberChef, Python.
- Ghidra, YARA and ClamAV for safe malware-analysis exercises.
- Ollama with a local model for AI exercises; no paid AI API is required.
- Trial products may be used where available, but trial-based software is not described as permanently free.

Safety & Ethics

- Only test systems you own or have explicit authorization to test.
- Use deliberately vulnerable VMs, synthetic credentials, synthetic logs and instructor-provided PCAPs/samples.
- No public-target scanning, unauthorized credential attacks or destructive activity.
- Wireless labs must use instructor-owned isolated equipment.
- AI-agent exercises use read-only tools unless a controlled action is explicitly approved; validate AI conclusions against evidence.

Assessment

Advanced Offensive Security Labs: 15%

SOC/SIEM Investigation: 20%

Detection Engineering & Threat Hunting: 15%

SOAR & Automation: 10%

AI Security & Agent Engineering: 15%

AI Analytics/Governance: 10%

Integrated Case Studies: 5%

Expert Capstone: 10%

Student Deliverables

- Rules-of-engagement and lab documentation

- Reconnaissance and network-enumeration reports
- Vulnerability/web/system security reports
- PCAP and malware/IOC analysis where applicable
- Splunk/Sentinel investigation and detection artifacts for Expert learners
- Threat-hunting/ATT&CK artifacts for Expert learners
- AI-assisted analysis or AI-agent project artifacts
- Final capstone report and presentation

Course Differentiation

This course is the advanced continuation of the Professional program. It assumes basic ethical-hacking competence and concentrates on engineering, operations, detection, response, automation and AI security rather than repeating introductory offensive-security material.

Reference Alignment

AI-security modules are informed by NIST AI Risk Management Framework guidance and the OWASP GenAI/LLM security project. These references support the curriculum but do not make the course an official certification.

Important Positioning Note

Independent professional training program. Not official EC-Council CEH courseware and not affiliated with EC-Council, Microsoft, Splunk, OWASP, Kali Linux or MITRE.

Recommended Delivery

120 hours total. Suggested format: 40 sessions × 3 hours, or 30 sessions × 4 hours, with substantial hands-on lab time.